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**SETP4583- SURFACTANT AND DETERGENT TECHNOLOGY**

**SECTION EB01**

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# 1.0 Introduction and Product Concept

## 1.1 Background

Liquid dishwasher detergents are a necessity in the house that is used to clean dishes like cutlery, crockery, and kitchen utensils off and eliminate oils, grease and food residues. The taste of the contemporary consumer market has changed the requirements of dishwashing liquids beyond the scope of cleaning effectiveness. The consumers are now looking at products which provide both the strong degree of degreasing function and skin safety such that high level of frequent usage does not cause skin irritation or dryness. Besides, the increased focus on environmental sustainability is pushing the necessity of creating formulations with biodegradable ingredients and a decreased amount of chemicals with no impact on performance.

The given report describes the formulation, preparation, and evaluation of Aqua Shine Dishwashing Liquid, a prototype water-based detergent that should address these new needs in the form of a scientifically balanced combination of surfactants.

## 1.2 Product Concept

The overall idea of Aqua Shine Dishwashing Liquid is to offer an effective cleaning product that is tough on Grease, mild on Hands. The formulation of the product is a homogeneous solution that is based on water, and this solution is made up of 88.5% water, thus making it a resource-saving and economical product.

The effectiveness of the product depends on the combined surfactant:

1. Cleaning Power: It applies anionic surfactants (SLES and LABS) in the deep cleaning and heavy grease removal.

2. User Experience: The formulation is formulated to give out a thick, stable foam, which is a visual sign of cleaning power to give it a visual indication of consumer satisfaction.

3. Safety and Mildness: The feature that makes the product concept stand out is the use of Cocamidopropyl Betaine (an amphoteric surfactant) that is used in large amount (4%). This is carefully incorporated in order to contain the severity of the anionic surfactants forming a gentle formulation on the skin.

4. Stability: Stability of the product is maintained by the presence of non-ionic surfactants that allow the product to be used in different temperatures and levels of hard water.

Aqua Shine is meant to be biodegradable and to be agreeable to normal wastewater disposal, and meets the underlying requirements of basic sustainability objectives without affecting the performance levels of a typical house cleaning detergent.

# 2.0 Ingredient List and Formulation Rationale

The formulation is water-based, comprising 88.5% of the total volume to act as the primary solvent.

## 2.1 Ingredient List

| Chemical | Type | Percentage (%) | Primary Function |
|---|---|---|---|
| SLES | Anionic Surfactant | 4% | Main cleanser and foaming agent |
| LABS | Anionic Surfactant | 1.5% | Heavy grease removal |
| Betaine | Amphoteric Surfactant | 4% | Foam stabilizer and skin protector |
| Berol | Non-ionic Surfactant | 2% | Grease solubilizer and stabilizer |
| Water | Solvent | 88.5% | Carrier for active ingredients |

## 2.2 Formulation Rationale

SLES (4%)

Chosen as the primary surfactant to provide the rich, stable foam consumers associate with cleaning power. At 4%, it balances effective detergency without being overly harsh.

LABS (1.5%)

A secondary surfactant that is stronger than SLES at removing oils and fats. It is used at lower percentage to minimize skin irritation while boosting degreasing efficiency.

Betaine (4%)

Added to reduce the irritation potential of the anionic surfactants such as SLES and LABS that are more irritative, making the product mild to skin. It also improves viscosity and ensures foam remains stable in greasy water.

Berol (2%)

A non-ionic surfactant that excels at emulsifying oils that anionics might struggle with. It ensures performance remains consistent across different temperatures and water hardness levels.

# 3.0 Preparation Method and Process

The preparation method and process will be discussed in 3.0.

## 3.1 Materials and Equipment

The ingredients used in the preparation of the liquid dishwashing detergent were SLES, LABS, Betaine, Berol and distilled water. These ingredients have been chosen on the basis of their surfactant characteristics and compatibility in aqueous formulation.

The equipment used in this experiment was kept simple and suitable for small-scale prototype development. A digital weighing balance was used to measure the required amount of each ingredient. Several plastic containers were used for mixing and storage of the detergent, while a wooden chopstick was used as a manual stirring tool to ensure uniform mixing of the formulation.

Despite the simplicity of the equipment, the preparation process was sufficient to produce a homogeneous detergent suitable for performance testing and evaluation.

## 3.2 Preparation Procedure

The liquid dishwashing detergent was made by a simple batch mixing process. The preparation procedures were done as follows:

1. Distilled water which constitutes 88.5% of the entire formulation was initially weighed and pour into a clean container at room temperature.
2. A small amount of Linear Alkylbenzene Sulfonate (LABS) was slowly incorporated into the water. Constant mixing was also ensured to provide consistency in dispersion and to make it more grease removable.
3. The Sodium Lauryl Ether Sulfate (SLES) was added into the water while manually stirred to ensure uniform dissolution and to minimize excessive foaming.

4. Betaine was then incorporated alongside the formulation to enhance the foam stability, minimize the irritation of the anionic surfactants and increase the general mildness.
5. To enhance the ability of oil emulsification and formulation stability in varying conditions of water hardness and temperature, a non-ionic surfactant, Berol, was introduced slowly into the mixture.
6. The solution was manually stirred using a wooden chopstick until the resulting mixture became clear and homogeneous.
7. A calibrated pH meter was used to determine the pH of the final product to make sure that it was within an appropriate range to be used in dishwashing.
8. The stocked detergent was left to rest briefly to release any air bubbles trapped in it and then the stocked detergent was transferred into a container where it was tested and assessed.

# 4.0 Test Results and Performance Evaluation

## 4.1 pH Test

The pH of the formulated dishwashing detergent was measured using a calibrated pH meter at room temperature. The pH was found to be within the recommended range of 6.5 to 8.5 for hand dishwashing liquids. This indicates that the formulation is mild enough for frequent skin contact while still providing effective grease removal.

## 4.2 Foaming Ability Test

The foaming capacity of the detergent was measured by placing a diluted sample in a container and measuring the amount and stability of the foam. The formulation resulted in a deep and stable foam, which lasted a reasonable period. This implies that there was a good surfactant activity, especially by SLES which is the main foaming agent, and Betaine helped in foams and lowered the foam collapse when grease was used.

## 4.3 Cleaning Performance

The cleaning performance of the detergent was assessed by applying it to greasy dishes followed by rinsing with water. The detergent effectively removed grease, forming a milky emulsion during washing. This observation demonstrates successful micelle formation, where the hydrophobic tails of the surfactants encapsulated oil and grease while the hydrophilic heads interacted with water, allowing the dirt to be rinsed away efficiently.

# 5.0 Biodegradables and Toxicity Assessment

## 5.1 Biodegradability of Ingredients

The dishwashing liquid formulation was designed using commonly applied surfactants with relatively favourable environmental profiles. The primary surfactants used include Sodium Lauryl Ether Sulfate (SLES), Linear Alkylbenzene Sulfonate (LABS), Cocamidopropyl Betaine (Betaine), and Berol (non-ionic surfactant).

SLES is known for its good biodegradability under aerobic conditions and is widely used in household cleaning products. LABS, when properly neutralized and formulated, is considered readily biodegradable and remains one of the most accepted anionic surfactants in the detergent industry. Cocamidopropyl Betaine is derived from fatty acids and exhibits high biodegradability with relatively low environmental persistence. Non-ionic surfactants such as Berol are also generally biodegradable and effective at low concentrations, reducing overall environmental load.

Water, which makes up most of the formulation, does not pose any environmental risk. Overall, the selected ingredients are expected to break down into simpler compounds through microbial action in wastewater treatment systems, minimizing long-term environmental impact.

## 5.2 Toxicity Considerations

The toxicity of the formulation was evaluated qualitatively based on known safety data of the raw materials and their typical usage levels. At the concentrations used, the surfactants are considered safe for household cleaning applications. SLES and LABS may cause mild irritation to skin or eyes at high concentrations; however, dilution during use and the presence of Betaine help reduce irritation potential.

Cocamidopropyl Betaine acts as a mild amphoteric surfactant that improves skin compatibility and reduces the harshness of anionic surfactants. The formulation does not contain heavy metals, solvents, or highly toxic preservatives, further lowering toxicity risks.

Proper labelling and instructions for use are still required to ensure user safety, particularly advising avoidance of direct eye contact and prolonged skin exposure.

# 6.0 Sustainability and Safety Considerations

## 6.1 Sustainability Aspects

Sustainability was considered during ingredient selection and formulation design. The surfactants used are widely available, cost-effective, and compatible with large-scale production, which reduces resource inefficiency. The high-water content minimizes the amount of active chemicals required while still maintaining effective cleaning performance.

Several ingredients, such as Cocamidopropyl Betaine, are derived from renewable resources (plant-based fatty acids), contributing to improved sustainability compared to purely petroleum-based chemicals. In addition, the biodegradability of the surfactants supports reduced environmental accumulation after disposal.

Future improvements could include the use of fully bio-based surfactants, eco-certified raw materials, or recyclable packaging to further enhance the product's sustainability profile.

## 6.2 User and Environmental Safety

From a safety perspective, the formulation is intended for routine household use. The product operates effectively at neutral to mildly alkaline pH, which reduces the risk of surface damage and skin irritation. The absence of strong acids, alkalis, or volatile organic solvents enhances user safety.

Safe handling practices include storing the product in a sealed container, keeping it out of reach of children, and avoiding ingestion. Environmentally, proper disposal through household wastewater systems ensures dilution and treatment before release into natural water bodies.

Overall, the formulation balances cleaning efficiency, user safety, and environmental responsibility, making it suitable for practical dishwashing applications.

# 7.0 Reflection on Team Collaboration and Learning

## 7.1 Team Collaboration

The design of Aqua Shine Dishwashing Liquid involved team work that had to be well synchronized with all the five members of the team. There was a clear division of task that made sure that the study of the properties of ingredients, preparation of the lab formulation and the later performance testing would be efficiently done.

The possibility to problem-solve on the formulation stage was one of the most important strengths of our cooperation. As an example, simple manual stirring took time and strict discipline on the part of the group to ensure that the mixture was rendered clear and uniform. There was also a need to have a mutual consensus of the Product Concept, which meant that we all had a common understanding that we were aiming to strike a balance between "Tough on Grease" and the Mild on Hands. The common vision assisted us in coming up with a unanimous decision on the ratio of the ingredients to be contained in the product, that high water content (88.5) to ensure that the product was cost effective and at the same time effective active ingredients to be contained.

## 7.2 Technical Learning Outcomes

This project was a priceless practical exposure to surfactant technology, the connection between the theory and practice. The Surfactant Synergy: We had an eye opener that as many as we can do with one surfactant is very little in terms of quality product. We could see with our eyes how addition of Anionic surfactants (SLES and LABS) to the cleaning power was combined with Amphoteric surfactant (Betaine) to form a synergistic effect. In particular, we got to know that besides being a foam stabilizer, Betaine plays a vital role in minimizing the skin irritation of harsher surfactants such as LABS. The Significance of Rational behind Formulation: It dawned on us that formulation is a balancing exercise. An example is that although LABS is an effective degreaser, we reduced it to 1.5 percent to avoid irritation of the skin, and we substituted it with 4 percent of SLES to provide foaming and detergency. This has helped us learn that more is not necessarily better, the ratio is what determines performance and safety. Process Awareness / Safety awareness: The order of addition was indicated as important during the preparation process. We were taught to add LABS gradually as a way of dispersion and also to include Berol so as to stabilise the formula with different water hardness. Furthermore, the importance of the safety testing in the development of a consumer product was supported by the pH test and the discovery of the product within the 6.5-8.5 range.

## 7.3 Conclusion

In general, this project has proven that the development of a good detergent must be based on a thorough understanding of the chemical reactions and the needs of the consumers. We managed to go beyond mere mixing to the explanation of why all the ingredients are there, including the cleaning properties of the surfactants used and the sustainability aspect of using biodegradable materials, such as Betaine and Berol. This experience has provided us with necessary skills in our future operations in designing chemical products.

# 8.0 Marketing Poster

Figure 5.1 shows the marketing poster for the Aqua Shine Dishwashing Liquid. The poster highlights the reason of choosing this product, instructions for use, precautions while using this product, sustainability claim, and the promotion provided.


Figure 8.1 Marketing Poster for Aqua Shine Dishwashing Liquid